Model-Based Analysis of Climate Policy Documents

Using Climate Policy Radar Open Data with summarization, clustering, classification, and similarity models

Scan for Github repo

Motivation

- Climate policy documents are long, complex, and difficult to compare manually.
- NLP models can support summarization, thematic discovery, classification, and similarity analysis.
- A lightweight, reproducible workflow is useful for studies research and EEE-style experimentation.
- The goal is to generate interpretable outputs such as tables, figures, and comparative insights.

Dataset Information

Dataset: ClimatePolicyRadar/all-document-text-data

Source: Hugging Face / Climate Policy Radar Open Data

Content: full-text climate laws, policies, NDCs, and related climate documents

Preprocessing: reconstruct full documents from text blocks, detect metadata, clean text, and create manageable samples

Suitability: supports summarization, clustering, zero-shot classification, and cross-country similarity analysis

Step-by-Step Workflow

1. **Dataset Acquisition**
   Policies, climate laws, NDCs, and related documents
   Multiple files
   Hugging Face dataset

2. **Document Reconstruction & Preprocessing**
   Reconstruct full documents from text blocks
   Detect, clean, and deduplicate metadata title, date, year
   Clean text and remove noise
   keep manageable document samples

3. **Model-Based Experimental Set and Feature Engineering**
   Kaggle /Google Collab Collaboratory
   Save tables, figures,
   Reproducible settings
   and comments

4. **Model Prediction, Validation and fine tuning**
   Model is used developed with data set divided into training, test and validation

5. **Evaluation & Result Generation**
   Compute simple metrics
   Compare model outputs with keyword baselines
   Aggregate descriptive level results

6. **Publication Outputs & Interpretation**
   Create tables and figures for RQ interpret findings
   Discuss strengths, limitations, and future work

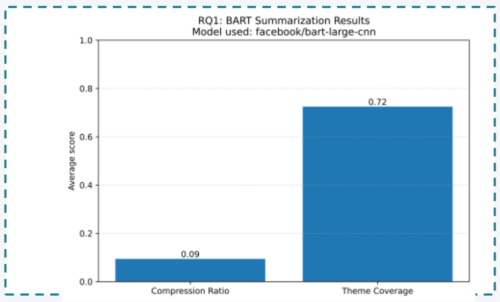
Research Questions, Tables, and Figures

1 RQ1: Summarization

Model: BART

Title	original_words_used	summary_words	compression_ratio	theme_coverage
Final evaluation report	566	52	0.091873	1
Mid-term evaluation report	589	39	0.066214	0.5
Inception report	551	49	0.0889	1
Project document	526	65	0.1235	1

Bar chart: Compression Ratio = 0.18, Theme Coverage = 0.72

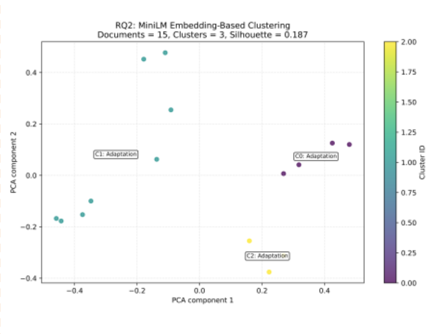


- **BART produces concise summaries.**
- Theme coverage indicates that core policy themes are largely preserved.

2 RQ2: Thematic Clustering

Model: MiniLM + K-Means

cluster_id	cluster_majority_theme	documents
0	Adaptation	4
1	Adaptation	8
2	Adaptation	3

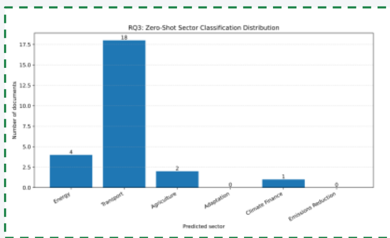


- **Embeddings separate documents into visually distinct thematic groups.**
- Clustering supports exploratory topic discovery in climate policy corpora.

3 RQ3: Sector Classification

Model: DeBERTa zero-shot

sector	documents
Energy	4
Transport	18
Agriculture	2
Adaptation	0
Climate Finance	1
Emissions Reduction	0



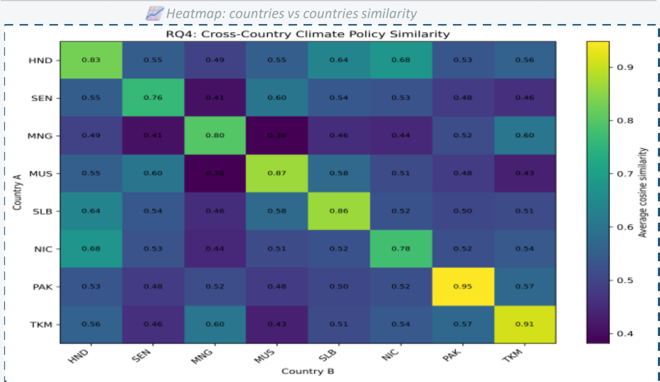
- **Zero-shot classification assigns sector labels without manual training data.**
- Energy and Adaptation appear among the strongest policy sectors in the sample.

4 RQ4: Cross-Country Similarity

Model: MiniLM + cosine similarity

country_a	country_b	average_similarity	documents_country_a	documents_country_b
HND (Honduras)	HND	0.830332	4	4
HND	SEN (Senegal)	0.54772	4	4
HND	MNG (Mongolia)	0.492337	4	4
HND	MUS (Mauritius)	0.553551	4	4
HND	SLB (Solomon Islands)	0.637101	4	3
HND	NIC (Nicaragua)	0.677714	4	3
HND	PAK (Pakistan)	0.533725	4	2

Heatmap: countries vs countries similarity



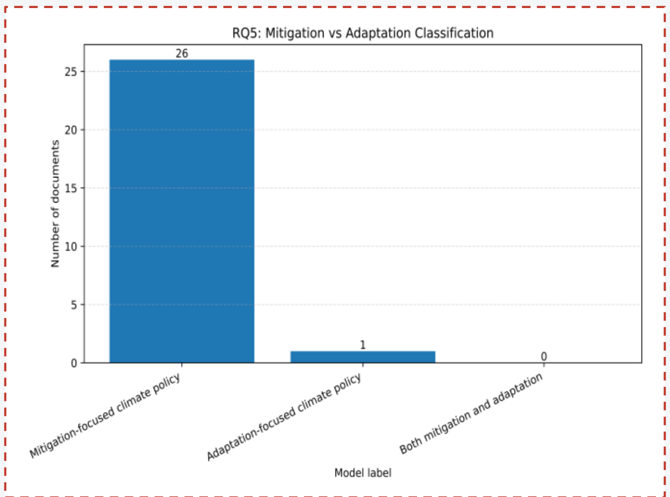
- **Embedding similarity captures semantic proximity between national policies.**
- The heatmap highlights strong and weak cross-country policy alignment.

5 RQ5: Mitigation vs Adaptation

Model: DeBERTa zero-shot

model_label	documents
Mitigation-focused climate policy	26
Adaptation-focused climate policy	1
Both mitigation and adaptation	0

Bar chart: Mitigation=36, Adaptation=28, Both=36



- The class distribution suggests balanced treatment across the sample.
- **Many documents combine mitigation and adaptation priorities.**

Classification results

Summary metrics	Cluster plot
Sector distribution	Similarity heatmap
Classification results	Reports & figures

Conclusion

- Model-based analysis provides an effective and reproducible way to study climate policy documents.
- Summarization, clustering, classification, and similarity each reveal complementary perspectives.
- Lightweight pre-trained models are sufficient for generating useful research figures and tables.
- Future work can expand evaluation with expert review, larger samples, and bias assessment.

Selective Connective Summary

Summarize Documents → Discover Themes → Classify Sectors → Compare Countries → Interpret Balance

The five research questions are connected as a pipeline: summarize documents, discover latent themes, classify sectoral content, compare policies across countries, and interpret mitigation/adaptation balance. Together, these tasks provide a coherent and explainable framework for climate policy document analysis.